Reflective Journal

Module 3 Lab | Trilok Skalani | ITAI1371

Getting Started

Honestly when I first opened the notebook I thought most of it would be straightforward since I had done some Python before. But once I actually started working through it I realized there were a few things I thought I understood that I actually did not. The biggest one was the train test split. I knew you were supposed to do it but I could not really explain why until I ran the models and saw the difference between training accuracy and test accuracy. That is when it clicked for me. A model can basically memorize all the training examples and score perfectly on them but that tells you nothing about how it will do on new data. The test set is the only honest measure of that.

What I Learned About the Three Types of ML

Before this lab I kept mixing up supervised and unsupervised learning. Now I think about it this way. Supervised learning is when you already know the answers for your training examples and you want the model to learn to predict those answers for new ones. That is exactly what we did with the wine dataset. We had the labels telling us which wine was which and the model learned from that. Unsupervised is when you do not have any labels and you just let the algorithm find patterns on its own, like grouping customers by behavior without knowing the groups ahead of time. Reinforcement learning is different from both because the model is not learning from a fixed dataset at all. It is an agent that takes actions and learns from whether those actions led to a good or bad outcome over time, like training a robot or a game playing AI.

Working Through the Workflow

The part that surprised me most was how much the EDA step mattered. I initially thought it was just making plots before the real work started. But when I looked at the correlation heatmap I could actually see which features were strongly related to the wine class. That directly helped me in Task 1 when I had to pick three features myself. I chose alcohol, color intensity, and proline and my model did better than the default feature set. That would not have happened if I had skipped the exploratory step.

The confusion matrix also took me a minute to fully understand. At first I just saw a grid of numbers. What helped was thinking about each cell as a specific question, like how many times did the model predict class 1 when it was actually class 2. Once I framed it that way the diagonal versus off diagonal values made a lot more sense. A perfect model would only have numbers on the diagonal because every prediction would match the true label.

Where This Connects to Real Life

The case study on fraud detection stood out to me. I had always thought of fraud detection as a rule based system but it makes much more sense as a supervised learning problem. You have thousands of past transactions labelled as fraud or not fraud and the model learns the patterns. What I did not consider before is that accuracy alone is a bad measure for that problem because fraud is so rare. A model that just says "not fraud" every single time would be 99.9 percent accurate and completely useless. That made me want to learn more about precision and recall and how to properly evaluate a model when the classes are imbalanced.

What I Still Want to Figure Out

A few things came up that I want to dig into more. I want to understand cross validation better because one train test split still feels a bit random to me depending on which samples end up in each set. I also want to know more about what happens when one class has way fewer examples than another and how you handle that properly. And I am curious about Random Forests since from what I read they are basically a bunch of decision trees working together, which sounds like it should be more reliable than one tree alone.